

Cătălin I. Cârstea

Associate Professor · Department of Applied Mathematics

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Personal information

Citizenship	Romanian
Date of birth	8 January 1980
Languages	Romanian (native); English (fluent); Mandarin (intermediate)

Education

2006-2010	Ph.D. in Mathematics · University of Chicago Advisers: Carlos Kenig and Wilhelm Schlag Thesis: <i>A construction of blow up solutions for co-rotational wave maps</i>
2004-2006	M.Sc. in Mathematics · University of Chicago
2003-2004	Study in Physics · University of Chicago, Department of Physics
2002-2003	Study in Physics · University of Bucharest, Faculty of Physics
1998-2002	Diplomă de Licență · University of Bucharest, Faculty of Physics

Research interests

Partial differential equations · Inverse problems

Employment

2023-present	Associate Professor · Department of Applied Mathematics National Yang Ming Chiao Tung University, Hsinchu, Taiwan
2018-2023	Specially Appointed Associate Researcher College of Mathematics, Sichuan University, Chengdu, China
2017-2018	Postdoctoral Fellow HKUST Jockey Club Institute for Advanced Study, Hong Kong
2015-2017	Postdoctoral Researcher National Center for Theoretical Sciences, National Taiwan University, Taipei
2014-2015	Lecturer in Physics · Brockenhurst College, Brockenhurst, UK Teaching at No. 7 Middle School, Chengdu, China
2012-2014	Staff Member · Rochester Zen Center, Rochester, NY, USA Repair and maintenance; bookkeeping.

2010-2012	Visiting Assistant Professor Department of Mathematics, University of Rochester, Rochester, NY, USA
2006-2010	Lecturer in the College Department of Mathematics, University of Chicago, Chicago, IL, USA
2005-2006	College Fellow Department of Mathematics, University of Chicago, Chicago, IL, USA
2003-2004	Teaching Assistant Department of Physics, University of Chicago, Chicago, IL, USA

Teaching

National Yang Ming Chiao Tung University

Calculus I and II; Introduction to Partial Differential Equations; Vector Calculus; Degenerate Elliptic and Parabolic PDEs; Colloquium.
Fall 2026: Calculus A (I); Introduction to Real Analysis.

Sichuan University

Introduction to Pseudodifferential Operators; Quantum Mechanics from a Mathematical Point of View.

University of Rochester

MTH 142 (Calculus II); MTH 143 (Calculus III, twice); MTH 162 (Calculus IIA, twice); MTH 161Q (Quest Calculus); MTH 240 (Introduction to Topology); MTH 266 (Real Analysis II / Fourier Analysis).

University of Chicago

Lecturer: Math 151-153 (Calculus, one year); Math 131-133 (Elementary Functions and Calculus, three years).
College Fellow: Math 161-162 (Honors Calculus); Math 204 (Analysis in $\mathbb{R}^n$).
Teaching Assistant: Physics 121-123 (General Physics).

Brockenhurst College

A-level and AP Physics for students in years 10-12 at the International High School of Jiaxiang Foreign Languages School, attached to No. 7 Middle School, Chengdu, China.

Grants

2024-2027	Inverse problems for elliptic and parabolic equations with anisotropy or quasilinear nonlinearity National Science and Technology Council (Taiwan) 113-2115-M-A49-018-MY3 Principal investigator · TWD 2,917,000
2023-2024	Inverse problems for equations featuring anisotropy, nonlinearity, or randomness National Science and Technology Council (Taiwan) 112-2115-M-A49-002 Principal investigator · TWD 745,000

Publications

1. Cătălin I. Cârstea and Ali Feizmohammadi. Two uniqueness results in the inverse boundary value problem for the weighted p-Laplace equation. *Forum of Mathematics, Sigma* 13 (2025), e147, 1-14.
2. Cătălin I. Cârstea, Tuhin Ghosh, and Gen Nakamura. An inverse boundary value problem for the inhomogeneous porous medium equation. *SIAM Journal on Applied Mathematics* 85(1) (2025), 278-293.
3. Cătălin I. Cârstea, Matti Lassas, Tony Liimatainen, and Lauri Oksanen. An inverse problem for the Riemannian minimal surface equation. *Journal of Differential Equations* 379 (2024), 626-648.
4. Cătălin I. Cârstea, Ali Feizmohammadi, and Lauri Oksanen. Remarks on the anisotropic Calderón problem. *Proceedings of the American Mathematical Society* 151(10) (2023), 4461-4473.
5. Cătălin I. Cârstea, Tuhin Ghosh, and Gunther Uhlmann. An inverse problem for the porous medium equation with partial data and a possibly singular absorption term. *SIAM Journal on Mathematical Analysis* 55(1) (2023), 162-185.
6. Cătălin I. Cârstea and Ali Feizmohammadi. A density property for tensor products of gradients of harmonic functions and applications. *Journal of Functional Analysis* 284(2) (2023), article 109740.
7. Cătălin I. Cârstea, Ali Feizmohammadi, Yavar Kian, Katya Krupchyk, and Gunther Uhlmann. The Calderón inverse problem for isotropic quasilinear conductivities. *Advances in Mathematics* 391 (2021), article 107956.
8. Cătălin I. Cârstea and Ali Feizmohammadi. An inverse boundary value problem for certain anisotropic quasilinear elliptic equations. *Journal of Differential Equations* 284 (2021), 318-349.
9. Cătălin I. Cârstea. On an inverse boundary value problem for a nonlinear time harmonic Maxwell system. *Journal of Inverse and Ill-Posed Problems* (2021).
10. Cătălin I. Cârstea and Manas Kar. Recovery of coefficients for a weighted p-Laplacian perturbed by a linear second order term. *Inverse Problems* 37(1) (2020), article 015013.
11. Cătălin I. Cârstea and Jenn-Nan Wang. Propagation of smallness for an elliptic PDE with piecewise Lipschitz coefficients. *Journal of Differential Equations* 268(12) (2020), 7609-7628.
12. Cătălin I. Cârstea, Gen Nakamura, and Lauri Oksanen. Uniqueness for the inverse boundary value problem of piecewise homogeneous anisotropic elasticity in the time domain. *Transactions of the American Mathematical Society* 373 (2020), 3423-3443.
13. Cătălin I. Cârstea, Tu Nguyen, and Jenn-Nan Wang. Uniqueness estimates for the general complex conductivity equation and their applications to inverse problems. *SIAM Journal on Mathematical Analysis* 52(1) (2020), 570-580.
14. Cătălin I. Cârstea, Gen Nakamura, and Manmohan Vashisth. Reconstruction for the coefficients of a quasilinear elliptic partial differential equation. *Applied Mathematics Letters* 98 (2019), 121-127.
15. Cătălin I. Cârstea, Naofumi Honda, and Gen Nakamura. Uniqueness in the inverse boundary value problem for piecewise homogeneous anisotropic elasticity. *SIAM Journal on Mathematical Analysis* 50(3) (2018), 3291-3302.
16. Sombuddha Bhattacharyya and Cătălin I. Cârstea. Optimal stability estimate in the inverse boundary value problem for periodic potentials with partial data. *Journal of Mathematical Analysis and Applications* 466 (2018), 642-654.

17. Cătălin I. Cârstea and Jenn-Nan Wang. Uniqueness for the two-dimensional Calderón problem with unbounded conductivity. *Annali della Scuola Normale Superiore di Pisa, Classe di Scienze* (5) XVIII (2018), 1459-1482.
18. Cătălin I. Cârstea and Jenn-Nan Wang. Size estimates for the inverse boundary value problems of isotropic elasticity and complex conductivity in 3D. *Inverse Problems* 33(12) (2017).
19. Cătălin I. Cârstea. A construction of blow up solutions for co-rotational wave maps. *Communications in Mathematical Physics* 300(2) (2010), 487-528.

Preprints

1. Cătălin I. Cârstea and Jenn-Nan Wang. Uniqueness for the inverse Schrödinger problem in the plane with L^p potentials, $p > 1$. *arXiv:2609.28046*
2. Cătălin I. Cârstea. Relative formalization in Isabelle/HOL of a result in inverse problems. *arXiv:2608.23659*
3. Cătălin I. Cârstea, Matti Lassas, Jinpeng Lu, Lauri Oksanen, and Ziyao Zhao. An inverse free boundary problem. *arXiv:2608.21153*
4. Cătălin I. Cârstea. On the Calderón problem with piecewise polynomial anisotropic conductivities and many-flat-face interfaces. *arXiv:2606.15977*
5. Cătălin I. Cârstea and Tuhin Ghosh. An inverse source problem for the Monge-Ampère equation from large boundary data. *arXiv:2606.07064*
6. Cătălin I. Cârstea. A note on several inverse problems with generally random coefficients. *arXiv:2605.20004*
7. Cătălin I. Cârstea. Hölder Stability from Exact Uniqueness for Finite-Dimensional Analytic Inverse Problems. *arXiv:2605.06354*
8. Cătălin I. Cârstea and Tuhin Ghosh. Inverse boundary value problems for certain doubly nonlinear parabolic and elliptic equations. *arXiv:2603.08297*
9. Cătălin I. Cârstea and Philipp Zimmermann. Reconstruction of coefficients in the double phase problem. *arXiv:2504.01691*
10. Cătălin I. Cârstea, Tony Liimatainen, and Leo Tzou. The Calderón problem on Riemannian surfaces and of minimal surfaces. *arXiv:2406.16944*
11. Cătălin I. Cârstea. Prescribed nonlinearity helps in an anisotropic Calderón-type problem. *arXiv:2406.14970*